

The emerging science of content labeling: Contextualizing social media content moderation

Garrett Morrow¹  | Briony Swire-Thompson²  | Jessica Montgomery Polny¹ |
Matthew Kopec¹  | John P. Wihbey³

¹Northeastern University Ethics Institute,
Boston, Massachusetts, USA

²Northeastern University Network Science
Institute, Boston, Massachusetts, USA

³Northeastern University College of Arts,
Media and Design School of Journalism
and Media Innovation, Boston,
Massachusetts, USA

Correspondence

Garrett Morrow, Northeastern University
Ethics Institute, 360 Huntington Avenue,
401 Renaissance Park, Boston, MA 02115,
USA.

Email: morrow.g@northeastern.edu

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Abstract

In the online information ecosystem, a content label is an attachment to a piece of content intended to contextualize that content for the viewer. Content labels are information about information, such as fact-checks or sensitive content warnings. Research into content labeling is nascent, but growing; researchers have made strides toward understanding labeling best practices to deal with issues such as disinformation, and misleading content that may affect everything from voting to health. To make this review tractable, we focus on compiling the literature that can contextualize labeling effects and consequences. This review summarizes the central labeling literature, highlights gaps for future research, discusses considerations for social media, and explores definitions toward a taxonomy. Specifically, this article discusses the particulars of content labels, their presentation, and the effects of various labels. The current literature can guide the usage of labels on social media platforms and inform public debate over platform moderation.

1 | INTRODUCTION

Every day, more than 100 billion messages are shared across Facebook's communications products (see Facebook Company Info, 2021), and every minute more than 500 hr of video is uploaded to YouTube (Frangoul, 2018). While a good deal of this information is benign, platforms need moderation methods to grapple with a substantial volume of problematic content across a variety of dimensions. In the context of online information and the web, content moderation is the organized governance of user-generated posts and behavior by information intermediaries and social media platforms, covering user activities of many kinds and encompassing an increasingly wide variety of interventions (Gillespie et al., 2020; Roberts, 2019). All social media platforms moderate content to varying degrees, even if it is done in a way that is opaque to users. A content label is a form of moderation where the platform attaches additional

information to a piece of content so that the viewer can better evaluate it. Content labeling is one tool that the platforms use to organize, moderate, and govern content, in addition to removal or algorithmic sorting. A content label may be, among other possible forms, a fact-check, a click-through barrier that provides a warning, a content sensitivity alert, or the provision of additional contextual information. When social media platform moderators consider putting labels on content with certain characteristics, those internal policymakers must also consider what the ultimate goals of the label are, how effective each type of label can be expected to be, and whether the labels might have any unintended consequences.

Platform moderators must contend with many ethical and normative issues, such as balancing concerns over limiting freedom of expression against the potential harms various pieces of content can inflict on other users or society at large. This tension compels social media companies to make a judgment about which values to

champion, but with the understanding that moderation practices may need to change with ongoing circumstances. If the norms and values that platforms settle upon conflict with the values of civil society, the platform's users, or the state, these platforms draw pressure to change their moderation practices. To date, social media companies have struggled to respond to these content moderation concerns with consistency and transparency. The result has been that platforms have been continually criticized for inconsistent and ad hoc policies, a pattern that some scholars have characterized as a “cycle of shocks and exceptions” (Ananny & Gillespie, 2016).

Unsurprisingly, it appears that users, platforms, and regulators want different things. Social media users have the conundrum of both wanting to encounter less misinformation *and* wanting to face less censorship on their own posts (Knight Foundation, 2020; Vogels et al., 2020). Social media platforms would rather they encounter less blowback against their content moderation policies (Douek, 2020). And regulators and policymakers would prefer that online content causes less harm and lead to a more informed citizenry (Bidar, 2021). Beyond the complications that these conflicting values pose, executing content moderation and labeling effectively is a complicated task, one which relies upon settling a wide range of empirical matters which are not yet fully understood, such as how much detail to provide in a fact-check, the most effective aesthetic qualities of a label, or what label design best encourages people to update their beliefs. Furthermore, it will require an interdisciplinary approach to understand how best to label as there are many socio-technical complexities that make the scientific aspects of content labeling a challenge.

This literature review attempts to lay some of the necessary groundwork for the science of content labeling. The authors are a group of scholars from a range of disciplines, including a political scientist, an experimental psychologist, two media scholars, and a philosopher, all of whom have been working on various projects aiming to inform how content labeling can be done better. Because of our multidisciplinary orientation, this review does not proceed from one disciplinary perspective. Rather it is an attempt to collect the central empirical work that has been done that we believe is most relevant to the question of how content labeling can be improved. The goal was to use our collective skills to present the literature in a way that will be most useful to the widest range of scholars working on a range of questions related to online content labeling.

In this review, we highlight the empirical labeling-related literature both to raise important considerations for social media platforms already engaged in content labeling and to illuminate various gaps that would make promising targets for future research. Although academic research that

bears specifically on the problem of social media content labeling is nascent, it is growing quickly. Many advances have already been made toward understanding labeling best practices to deal with misinformation, conspiracy theories, and misleading content—aspects of our online information ecosystems that may affect everything from voting behavior to personal health to emergency response. However, we contend that the specific problem of how to execute content labeling—with what justification, with what tools and techniques, and for what ultimate purpose—is a problem that demands increasing definition and focus.

The following sections focus on various factors social media moderators should consider when labeling content. We begin with a discussion regarding information labeling in adjacent industries, and the history of content moderation on social media platforms. We next explore the state of veracity and context labeling, their aesthetic qualities, and current scholarly understanding of psychological effects. Some of these effects may be leveraged to create certain outcomes for platform users, and such strategies connect with the broad “nudge” literature that has proven influential in public policy and economics (Lorenz-Spreen et al., 2020; Thaler & Sunstein, 2009). We begin by putting content labeling into context by looking at the history of content moderation and informational labeling broadly. We then discuss the current state of content labeling where we consider current labeling practices. Next, we move on to the unintended effects of labels, and we conclude with a discussion on the future of labeling as a moderation practice.

2 | LABELS AND WARNINGS IN CONTEXT

Labeling is not a new practice, nor has content labeling suddenly emerged from social media into society. Information labeling in adjacent fields and industries provides a foundation for content labeling practitioners, and we foreground that research to build context for the present discussion of informational content labeling online. Furthermore, the current state of online content labeling has been informed by the history of online content moderation on platforms such as Facebook and Twitter. In this section, we put content labeling into its interdisciplinary context to situate our later discussion of the current state, unintended consequences, and future of online social media content labeling.

2.1 | Information labeling

Although content labeling online is a relatively recent phenomenon, information labeling more generally is an

old technique. Given that researchers have long been examining the efficacy of labeling strategies in various industries, arguably social media platforms could draw upon those related bodies of literature to inform their own labeling strategies. For example, library scientists have been thinking about the ethics of labeling since at least the late 1980s (Nesta & Blanke, 1991; Pendergrast, 1988; Wolkoff, 1996). Labeling discussions also occur in more traditional, transactional marketplaces. For example, the food industry has a rich literature investigating the efficacy of labels (Messer et al., 2017). Social media moderation researchers could draw upon food labeling literature such as research into genetically modified organism labels (Yeh et al., 2019), calorie labels (Golman et al., 2017; Thunström, 2019; Thunström et al., 2016), or nutrition (Bushman, 1998; Talati et al., 2019). Indeed, many phenomena that we have been observing with online misinformation have also been found with food labels (for instance, the implied truth effect; see Yeh et al., 2019). As another example, media warning labels for movies (e.g., Motion Picture Association of America's ratings), television, or music is an area of previous labeling research that online moderation researchers and practitioners might be able to gain useful insights from (Bushman, 2006; Bushman & Cantor, 2003; Bushman & Stack, 1996; Gosselt et al., 2012).

Industry-specific information labeling literature can be paired with the broader warning and security label research. Whereas information labeling on products predominantly aim to provide consumers with the knowledge needed to make informed purchasing decisions, a warning label is a more direct communication about the specific harms of a product (Wogalter & Laughery, 1996). Some examples are the labels that warn consumers of the harms of smoking or alcohol that come attached to cigarettes or bottles of wine. Warning labels can also take the form of hazard communication to inform behavior beyond commerce, for example, carbon monoxide warnings on propane stoves. Warning label researchers have used the Communication-Human Information Processing (C-HIP) model where there are four stages of information processing between a warning and a corresponding behavior (Conzola & Wogalter, 2001). A warning must (1) attract attention, (2) be comprehended by the viewer, (3) convince the viewer of the danger, and (4) motivate the viewer to change their behavior. A comprehensive literature review (Stewart & Martin, 1994) and multiple meta-analyses over the years (Argo & Main, 2004; Hancock et al., 2020) about warning labels finds that they are best for informing consumers, but not necessarily persuading or changing the behavior of consumers. People are selectively attentive to warning labels, as with information labels such as calorie counts, and a

variety of design considerations can alter the effects of different labels. While not every aspect of product labels or warning labels apply to social media content, platform moderation practices can be better informed when anchored by the empirical literature from these adjacent industries.

2.2 | History of content moderation on social media

The practice of content moderation on the social web began in earnest and at scale well after the founding of platforms such as Facebook (2004), YouTube (2005), and Twitter (2006). Moderation has significantly evolved with the growth and complexity of the platforms and has been characterized by what may be seen historically as several different phases. In their early years, social media companies such as Facebook and YouTube had relatively small teams devoted to content review, and moderation rules and policies were very limited in scope (Klonick, 2018). Even as platforms tightened and clarified their terms of service, policing of content was intermittent at best. As time passed, however, platforms began to remove content more actively, often in response to takedown requests from governments, from other third parties, or from public pressure relating to violent or offensive speech. Eventually, automated systems for flagging content, driven by machine learning techniques, were developed and became widely used to aid these moderation functions for companies, facilitating tactics such as filtering and downranking (making content less visible to users) of content (Bloch-Wehba, 2020).

In 2016, Facebook began partnering with third-party fact-checkers at media outlets, heralding the beginning of more active, but also more nuanced, moderation strategies (Silverman, 2016). Subsequently, many companies have begun using techniques other than removal or reducing visibility through algorithmic downranking. The application of contextual, corrective, warning, source, transparency, or other kinds of labels has become more common, making it an emerging area of focus for companies and researchers alike (see Polny & Wihbey, 2021 for examples of platform-specific content labeling). Companies have applied labels either to create “friction” to limit the viral sharing and reach of misinformation or to warn or empower users through targeted informational disclosures (see examples of company policy: Facebook Community Standards, last visited December 23, 2020, and Twitter Rules, last visited December 23, 2020).

The nature and role of labeling among the suite of content moderation tools have become the topic of mainstream public debate. The debate over labeling practices

has been closely intertwined with discussions of reforming Section 230 of the Communications Decency Act, which has thus far effectively protected social media platforms from liability lawsuits within the U.S. legal system. For example, on May 29, 2020, Twitter labeled a Tweet about the George Floyd protests by the former U.S. President Donald Trump as “glorifying violence,” but ultimately deemed that retaining the Tweet on the platform was in the public interest, and therefore did not remove it. Twitter’s defense of the choice focused attention on Section 230, including an Executive Order targeting the provision (Dwoskin, 2020). Later in 2020, in response to a series of election-related Tweets by the former President, Twitter once again chose labeling over removal, this time attaching fact checks to the Tweets (Conger, 2020). Fears that disinformation might either be used to support electoral malfeasance in the 2020 U.S. presidential election or might exacerbate the outbreak of COVID-19, in turn, spurred social media companies to take on increasingly aggressive roles as arbiters of truth in the public sphere, all the while protected by Section 230. In the context of internet history, content labeling, with all the promises and perils it entails, has very much found its moment. Yet well-grounded, scientific principles must be applied when addressing these emerging problems. The available research literature that could guide these practices and their interpretation has not been coherently consolidated, and many areas remain in a relative state of confusion.

3 | THE CURRENT STATE OF CONTENT LABELING PRACTICE

Content labels are information about content that indicates a quality of the information. There are a variety of qualities that platform moderators could want to inform users about, therefore we can distinguish between different types of labels. Informational labels may be providing information about the specificities of a claim in the piece of content or, instead, the label may be more broadly contextual without addressing specificities of a claim. A common label is a veracity label, or fact-check. An example of a non-claim-specific label is a link to expert sources such as the CDC on COVID-19 related discussion. Claim-agnostic labels can vary in the type of contextual information provided. For instance, a YouTube content provider may have a contextual source label (e.g., “BBC is a British public broadcast service”), a Tweet could have metadata statistics on its origin or sharing trends, or a Facebook post that suddenly goes viral may have a simple reminder to viewers to engage in “critical thinking.” We thus turn to two common types of content labeling,

veracity labeling, and contextual labeling, before discussing labeling in the diverse social media platform landscape.

3.1 | Veracity labeling

Veracity labels seek to indicate the truth status of information that may be disputed, affirmed, or actively under review and can be negative, positive, or neutral, as with warning labels based on fact-checks that call a piece of content “misleading,” “truthful,” or “not verified.” Fact-checks and contextual information are used to refute and slow down the spread of rumors, misinformation, and disinformation. Where misinformation can be considered an umbrella term for all inaccurate information, disinformation specifies that the inaccurate information is spread with intent (Andersen & Sør, 2020; Fetzer, 2004; Rid, 2020; Wardle, 2017). Individuals or groups are purveying disinformation if they intend to deceive by spreading misinformation that they know to be false. No labels currently dissociate between misinformation and disinformation, as the intent is extremely difficult to determine and quantify (Aikin, 2013; Phillips & Milner, 2020). Veracity labels are particularly important given that false information has been shown to, on average, travel faster, farther, and deeper in social networks than true information (Vosoughi et al., 2018). Figure 1 shows an example of a false information label on Facebook content with a third-party fact-check and a click-through barrier.

While veracity labels might help to reduce belief in misinformation, users may still share misinformation for reasons other than providing other users with reliable informing (e.g., sharing for social, emotional, or political reasons), and such behavior may be more directionally motivated (Kunda, 1990). Pennycook et al. (2020) found that when people see political misinformation labeled as false, the intention to share the content reduces (also see Mena, 2020). One explanation for why labeling may slow sharing is that people are generally not explicitly considering veracity while engaging with social media, so the labels can help slow people down and make them think about what they are reading and potentially sharing with others (Pennycook & Rand, 2019b, 2020).

However, both Mena (2020) and Pennycook and Rand (2020) found that flagging a misinformation story does not completely eliminate intentions to share at the aggregate level. This is likely because people use social media and share content on social media for numerous reasons other than to factually inform other users. For example, many people will use social media and share to socialize or engage in self-expression (Chen et al., 2015; Oh & Syn, 2015), or because they think the content is

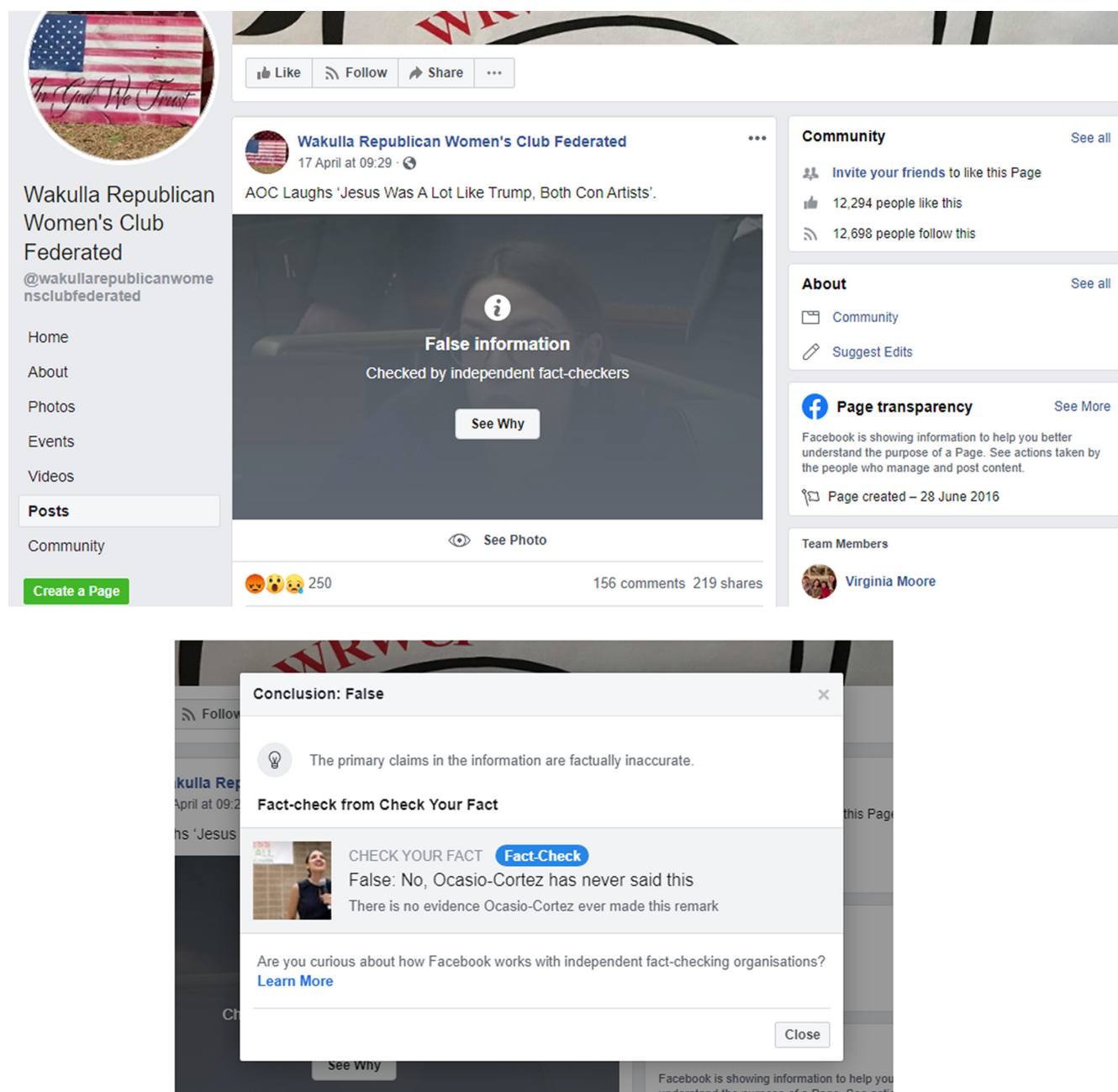


FIGURE 1 Third-party fact-check and a click-through barrier on Facebook

humorous or satirical. In short, while labeling may slow the velocity at which false content spreads on social media by correcting the misinformed beliefs it gives rise to and by reducing intentions to share it, there is more to be done to investigate the extent to which a label mitigates the spread of misinformation.

3.1.1 | Correction detail

One possibly important consideration is exactly how much detail should be in a label. On one end of the

spectrum corrections may simply include a “false tag” to indicate that information is inaccurate; on the other end are much more detailed approaches that inform individuals with several sentences or paragraphs about why the information has been deemed false. A recent meta-analysis showed that detailed corrections assisted belief updated in comparison to the misinformation being presented alone (Chan et al., 2017).¹ Studies not included in the meta-analysis that specifically manipulated detail also found that more detailed messages facilitated belief change, particularly after a delay (Ecker et al., 2020; Swire, Ecker, & Lewandowsky, 2017). Indeed,

according to the “Debunking Handbook” (Lewandowsky et al., 2020, p. 13), a detailed correction helps because the detail explains why the misinformation is incorrect, and why the alternative information is true. The detail allows people to correct their own inconsistencies (Ecker et al., 2017) and promote longer-lasting belief change (Ecker et al., 2020; Rich & Zaragoza, 2020).

Veracity content labels often have a lasting positive impression on the content consumer. In addition to Swire, Ecker, and Lewandowsky's (2017) findings cited previously, Ecker et al. (2020) tested a simple “false tag” retraction versus a short (Twitter-length 140 characters), more detailed refutation. Participants were first exposed to a piece of misinformation then a retraction or refutation to that misinformation. After a delay of either 1 day or 1 week, the participants were then tested to measure their belief in the initial misinformation. Ecker and his colleagues found that after just 1 day both correction types were effective in correcting people's misconceptions. However, after a week the simple retraction format was not as effective at correcting misinformation belief as the more detailed refutation. Yet, the detailed refutation format yielded sustained belief change. One remaining consideration regarding ideal levels of detail for veracity labeling is whether people are more likely to read short rather than long labels. Even if people are more likely to change their belief with detailed correction, if they are less likely to attend to them in real-world contexts, this would be a serious hindrance. After all, social media users are rarely incentivized to read lengthy labels in the ways that they tend to be incentivized in experimental psychology experiments. [Correction added on April 04, 2022, after first online publication: Repeated sentences were removed from this paragraph.]

3.1.2 | Correction source

Another component of content labeling is the source of the label, and the perceived credibility of that source by the content consumer is relevant to a label's effectiveness. One study found that under certain circumstances, corrections from “unlikely sources” (i.e., a person making corrections contrary to their interests) can help overturn politically motivated misperceptions (Berinsky, 2017). The author found that a Republican governor or chief executive of a fossil fuel company was better at correcting misperceptions about climate change held by Republican-leaning users. Unfortunately, due to our fractured media environment and increased political polarization, it would be rather difficult for content moderators to go around and find some trusted partisan politician for every piece of politically inconvenient misinformation that circulates.

While Berinsky's (2017) finding was not replicated by Swire, Berinsky et al. (2017), the authors hypothesized that this was potentially because their corrections were not sufficiently in opposition to the participants' core values. Furthermore, their sources were not attributed to real-world sources, but simply from a Democratic source, Republican source, and non-partisan fact-checking source.

3.2 | Contextual labeling

In contrast to veracity labels, contextual labels provide information to the user that may be missing from the user-generated post. Lorenz-Spreen et al. (2020) provide examples of how behavioral nudges and boosts (some of which are manifested as contextual information labels) can be used to affect user behavior. In short, a nudge intervention requires little active user engagement, while a boost requires some cognitive or motivational involvement from the user. Some examples of potential nudges and boosts for contextual labeling include: providing more information about how recent a shared story is; providing information on which other parties have been promoting a story and their relationship to the content viewer; providing information on why a content viewer may be seeing a specific piece of content; or telling a content viewer (or sharer) how many other people have been exposed to the same story.

3.2.1 | Source labels

In a sense, contextual labeling often overlaps with veracity labeling in those cases where the label provides more detail about the organization or author that created the news story or other piece of content. For example, rather than labeling the specific content of a reposted news story, the original source (e.g., news outlet or webpage) that originally authored the story could be labeled instead. Arnold et al. (2021) found that the use of source labels (e.g., “BBC is a British public broadcast service” in Figure 2 below) may help reduce belief in international misinformation and even mitigate intent to share, though the effects vary. Using ideologically congruent memes (in both Facebook and Twitter formats) labeled as “a foreign government account,” “a Russian government account,” or no source label at all, the authors showed that both self-identified Democrats and Republicans generally think the labeled posts are less truthful than the unlabeled posts. However, there appears to be some variation across social media platforms as labels on Twitter were found to be more effective than labels on Facebook in terms of both reducing truthfulness ratings and

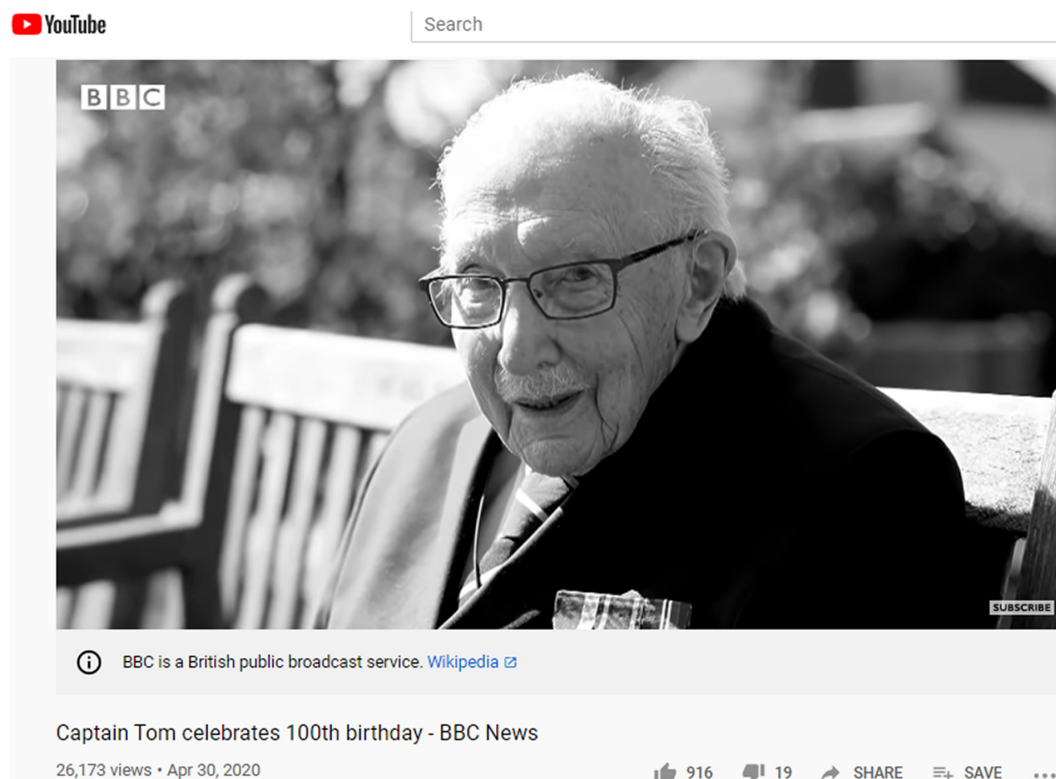


FIGURE 2 Contextual information panel with information about the content provider

reducing willingness to share. Nassetta and Gross (2020) find similar results in a study of YouTube state media labels, noting that platforms need to make sure that the labels are clear and non-subtle.

An alternative to source labels that provide context regarding the entity that originally generated the content, platforms could also use source ratings that provide an evaluation of the overall credibility of the source, without evaluating the truthfulness of any specific piece of content. Kim et al. (2019) found that low source ratings (i.e., 1 out of 5 stars on a news source) decrease believability and affect the way participants interact (e.g., comment or share) with a post overall, but high source ratings have little effect on either believability or user interaction. Additionally, Kim et al. (2019) and Kim and Dennis (2019) suggest that by emphasizing sources and their overall truthfulness ratings, users can become suspicious of any source without a rating, which could be either beneficial or detrimental to an overall labeling strategy, depending on the overall goals of the moderation policy. And both studies by Kim and colleagues found high levels of confirmation bias—where people are more likely to engage with ideologically congruent content regardless of content source ratings—which suggests that the widespread use of source labels that rate overall truthfulness may lead to unintended consequences.

Dias et al. (2020) found that visually emphasizing sources may not be effective at countering misinformation. The authors found that emphasizing source information without also supplying an overall truthfulness rating to users had no effect on content belief or interaction regardless of if the article was true or false, or if the article was from a known misinformation purveyor or mainstream source. Dias et al.'s experiments were followed by three additional surveys where people were asked to rate the trustworthiness and accuracy of headlines and sources. The authors found a strong positive correlation between headline plausibility and source trustworthiness. Therefore, Dias et al. concluded that adding information about the headline source did not add much additional information to participant's headline trust evaluations beyond what was already contained in the headline. The authors then follow-up these surveys with one last experiment where participants were exposed to "mismatched" articles (i.e., a plausible headline from a misinformation source) and found that providing information on the source increased the probability that participants would rate a true headline as false (and vice versa). In other words, emphasizing sources without also providing source ratings may be counterproductive.

The Knight Foundation and Gallup (2018) have also done research into the importance of attributing the

sources of news articles, finding that sources that tend to be perceived as having partisan agendas will often be rated as less trustworthy if the source is properly attributed than they are without the attribution. But the researchers also found that the perceived partisanship of the source had differential effects on participant beliefs and behaviors, depending on whether the partisan orientation of the source aligned with the orientation of the participant or not. This suggests that platforms that use source attribution as part of their content labeling strategy may need to take special care over how they label sources that are perceived as partisan, if they are operating in a highly polarized, politically partisan online environment.

3.2.2 | Virality measures

One barrier to the widespread use of claim-specific labeling such as fact-checking is the problem of scale. With millions of users posting new content and sharing existing content around the clock, some labeling strategies are easily scalable, some are scalable only with the assistance of artificial intelligence and machine learning technologies, and other strategies are more difficult to scale. The advantage of contextual labeling is that it is not as claim-specific as a fact check and can be scaled more rapidly than claim-specific fact checking. Providing source information or content metadata can surface and draw attention to data that the platforms (should) already have (see Lorenz-Spreen et al., 2020). For example, the platforms, in addition to basic like, comment, and share data, should have data on (1) when a piece of content was first posted, (2) what other URLs are cited in the piece of content, (3) how many other accounts promoted the piece of content, and (4) cascade statistics (Lorenz-Spreen et al., 2020).

Even when scalable, context labeling may run into other issues, such as platform interoperability or the affordance that media from one platform can be shared on a different platform. In interoperable platforms, labels may not carry over. For example, if someone watches a YouTube video about COVID-19 mask wearing on the YouTube platform, there is a label directing viewers to the CDC for up-to-date information. However, if someone were to watch the same YouTube video about COVID-19 but embedded in Facebook through someone sharing the video rather than through the YouTube URL or application, that CDC informational label may not be carried over.

The difficulty of scaling means that efforts to scale moderation will increasingly be automated (Bloch-Wehba, 2020). A number of media scholars have raised

worries about problems related to increased automation of moderation such as collateral censorship, input data issues, and a lack of transparency (Balkin, 2018; Balkin, 2020; Caplan, 2018; Gillespie, 2020). There is also the issue of “identical content in parallel context” (Bickert, 2021) where multiple people may post the same photo. If the context of the shared photo is similar to a post that has already been moderated, the post will get the same moderation treatment. However, if the context is different, for example, if the context condemns the moderated photo instead of supporting it, this would not be considered parallel (see Bickert, 2021 for more on Facebook’s view of this issue). Facebook claims that the difficulty in moderating “identical content in parallel context” is that determining context is difficult at scale.

3.3 | Labeling in the diverse social media platform landscape

Each social media platform has unique affordances, or actions that the technology allows for users (Phillips & Milner, 2020; Wellman, 2001), and therefore the different platforms require different approaches to content labeling and moderation. Such platforms include video players such as YouTube or Vimeo and mobile instant messengers such as WhatsApp or Signal. Currently, on YouTube (or videos embedded in other services), content labels are done primarily through textual labels that appear juxtaposed to a video, which are often contextual and informational. Advertising research has found that users are less likely to skip pre-roll advertisements that appeal to their interests (Campbell et al., 2017) and are non-intrusive (Goodrich et al., 2015). Research suggests that video-oriented social media might be able to effectively leverage appealing and non-intrusive pre-roll, PSA-style videos to provide contextual information similar to a label (perhaps in addition to the other label types highlighted in this review). In other words, by playing an informational video before a YouTube video, the content moderators could convey information to the viewer in a much richer mode than merely a textual label. Whether such videos would be effective as a moderation strategy is an open question, but there is some evidence from movie theater advertising research that suggests they could be effective (Ewing et al., 2001; Johnson, 1981; Wilson & Till, 2013).

Advertising research provides additional sources that can inform labeling practice, specifically labeling of sponsored content or influencers. In general, people are poor at recognizing sponsored content (Amazeen & Wojdyski, 2020; Wojdyski, 2016). Sponsored content is content paid for by an advertiser that looks like a normal piece of news or opinion from an outlet but is really a

piece of advertising for the company paying for the sponsorship (Krouwer et al., 2017). There are a variety of presentation characteristics, including informational label disclosures, that can better inform people about sponsored content (Boerman & van Reijmersdal, 2016; Krouwer et al., 2020). Indeed, Facebook has been experimenting with a variety of labeling formats for sponsored content on their platform (Sochurkova, 2018). Advertising researchers have also found that pre-roll videos can help promote transparency of influencer sponsorship (Evans et al., 2018). The ability to distinguish between hard news and opinion is important in informing the public, and the research into video advertising (Duffett, 2020) and influencer research (De Veirman et al., 2019) could help guide future labeling of videos, including combining labels with moderation videos.

Platforms that take advantage of other affordances, such as direct peer-to-peer communication apps such as WhatsApp may require other forms of labeling to moderate misinformation. WhatsApp is an especially popular mobile instant messaging tool in developing countries such as Nigeria, India, or Pakistan, and there is some evidence that these kinds of apps are increasingly implicated in the spread of misinformation. Many peer-to-peer communication applications do have some existing labels such as a “forwarded” label (Figure 3) and a “suspicious link” label (Supporting Information) but other labels may

be necessary depending on cultural context. For example, WhatsApp can be used for standard text messaging services, but also has other usages depending upon how people use the application in their everyday life. In countries without widespread, reliable internet connectivity such as India, Nigeria, and Pakistan, people send recorded audio messages to one another on WhatsApp and then listen to them as they travel from home to work to social locations (Pasquetto et al., 2020). Given that people download and listen to audio messages, “labels” to add contextual information may have to take different forms that are appropriate for audio messages, such as inoculation warnings (Guess et al., 2020) or virality data (Lorenz-Spreen et al., 2020) that account for the fact that audio does not allow information to be overlaid on other information (i.e., you cannot put the equivalent of a warning sign over sound).

Due to the closer peer-to-peer connections, misinformation can rapidly spread across groups via these messaging tools. Indeed, research on COVID-19 misinformation found that a higher belief in misinformation correlated with a higher usage of apps such as WhatsApp (Baum et al., 2020). However, the same social connectivity may also allow for potential social corrections (Rossini et al., 2021). In their study of WhatsApp usage, Pasquetto, Jahani, Baranovsky, and Baum (2020) found that the best source of mobile

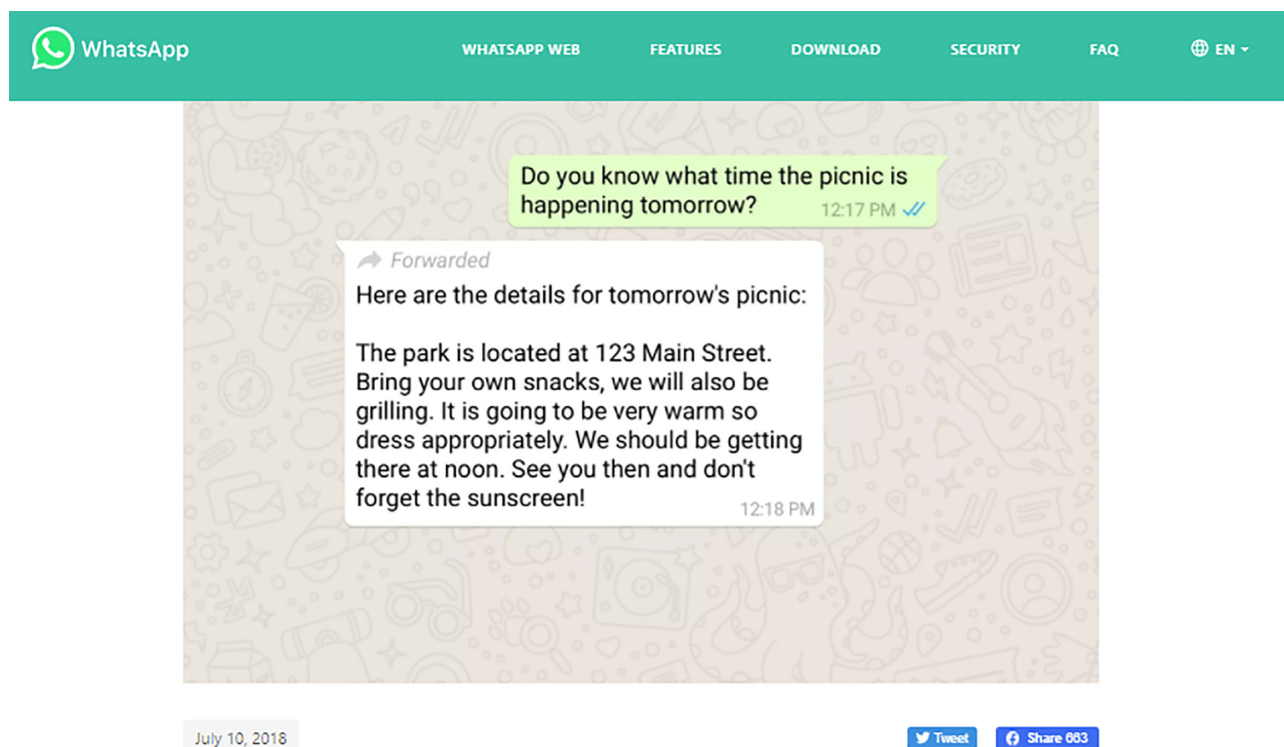


FIGURE 3 Context label indicating forwarded content on WhatsApp

messaging correction was like-minded family or friends, but that only highly-motivated people would engage in corrections. WhatsApp could encourage or reward corrections, and provide more contextual information via labels to lower the cognitive tax of issuing a correction.

3.4 | Graphics and design

Social media is predominately a visually oriented medium, meaning that many of the information exchanges that occur on the platforms are not purely textual. On these large socio-technical platforms, even small design tweaks to how information is presented or to how users interact can, at scale, have substantial downstream consequences. Color choices, visual hierarchies (the relative size and placement, and degree of prominence, of certain features versus others), and overall design architectures can matter significantly in terms of how users might choose to engage with and share content (Goodman et al., 2021; Mathur et al., 2019).

The effects of graphical elements are often ignored in the social media content moderation literature, but the warning label literature provides a wealth of useful lessons for moderators to draw upon. If the idea is to warn someone about something potentially harmful, like in Figure 4 from Reddit, then graphic designers should use red as a color because red has the most impact on hazard perceptions, followed by orange and yellow in declining levels of effectiveness (Chapanis, 1994). Various pictograms (such as the triangle with exclamation mark in

Figure 4) may also be effective in boosting the effects of the written warning reasoning (Jaynes & Boles, 1990; Riley et al., 1982). Additionally, the size, shape, and placement of warning iconography are also important (Hancock et al., 2020). There is a trade-off between how noticeable a warning label is (its intrusiveness) and viewer recall of the warning (its potential effectiveness).

In a 2021 study of interstitial warnings—for example, a warning screen between a google search and the content a user clicks on—Kaiser et al. found that such warnings can be more effective at conveying information about the risks of harmful content than contextual labels. However, the study did not find that interstitial warnings had any more of an effect on behavioral outcomes than contextual labels. Additionally, the use of small warning iconography and subtle text on interstitial warning screens lessened label effectiveness, while the use of bright red backgrounds increased their effectiveness. Two potential limitations of using interstitial warnings highlighted by Kaiser et al. are (1) the possibility that social media platforms will not want to use interstitial warnings with the optimal level of intrusiveness and (2) that an increased use of interstitial warnings may cause warning “fatigue” where they lose effectiveness over time. Currently, most of the interstitial warnings internet users encounter are those that flag potential cybersecurity threats, for example, when a user attempts to navigate to a page with an outdated security certificate. Additionally, repeated exposures to interstitial warnings surrounding misinformation may have detrimental effects on the overall power of such security

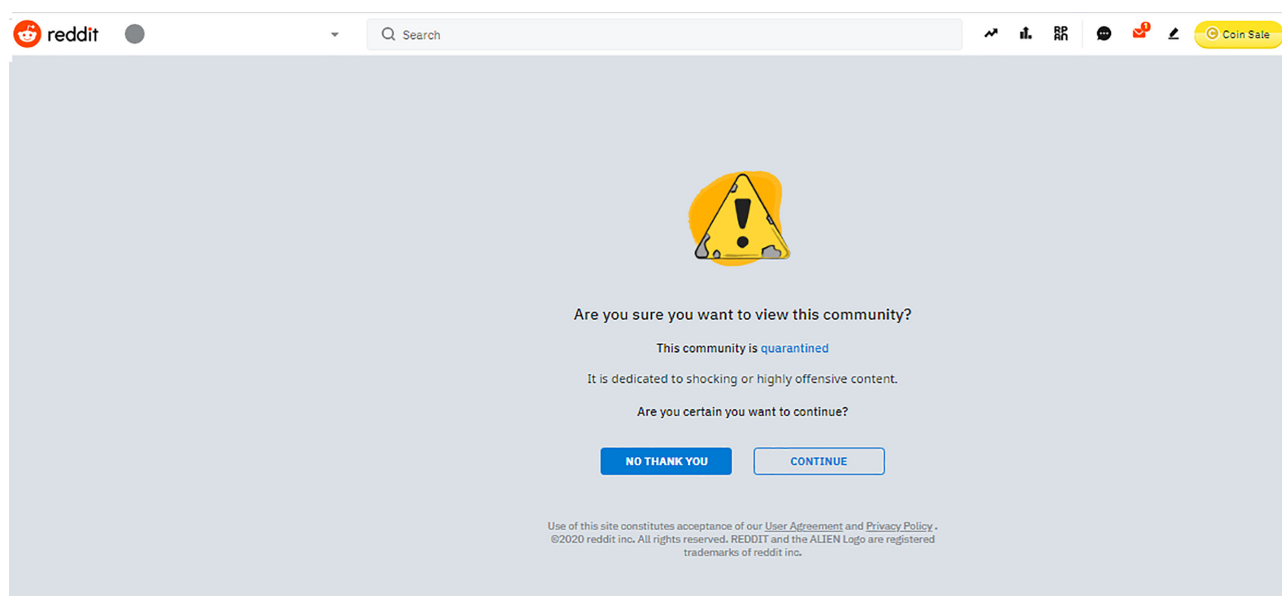


FIGURE 4 Quarantine Notice on Reddit. Note the coloring of the exclamation mark warning image and the shading of the “no thank you” and “continue” buttons

warnings. The best way to balance these concerns may be to simply draw upon the design lessons and graphical effectiveness of warning labels to improve contextual labels, as opposed to switching over to interstitial only strategies.

Social media users leverage the platforms to exchange ideas, discuss life, and socialize via posts, photos, memes, and videos. Sometimes content appears with text, sometimes without. The visual nature of many platforms suggests that purely textual labels may not be the best format for content moderation. A common graphical feature in labels (especially fact-checks) is the so-called “truth scale,” such as PolitiFact’s “Truth-O-Meter” that ranks a politician’s statement on a scale from true to “pants on fire.”² However, the efficacy of such graphical features is under debate. A recent meta-analysis (Walter et al., 2020) found that corrections that use graphical features such as a truth scale are not as effective as corrections that do not use a graphical element. However, the studies used in the meta-analysis did not manipulate graphical features themselves. Contrary to Walter et al.’s meta-analysis, Amazeen et al. (2018) found that under certain conditions, truth scales may enhance the efficacy of fact checks. For instance, they found that truth scales are more effective for correcting non-political information and lose efficacy the more politicized an issue is. Furthermore, Nyhan and Reifler (2019) found that graphics (e.g., charts) may be useful for presenting data and changing misperceptions, even if they do not actually improve understanding of the underlying facts. Using graphical elements is an area of research that needs attention, especially given the visual nature of social media platforms. [Correction added on April 04, 2022, after first online publication: Reference citation for Chan et al. 2017 was removed from this paragraph.]

Labels could take the form of systematic symbology that can take advantage of the warning label and media label literature. For example, Gosselt, De Jong, and Van Hoof (2012) and Bushman (2006) found that a standardized symbology of labels (e.g., the Motion Picture Association of America’s movie rating system or Newsguard’s news source credibility rating system) may affect views of the content therein. Video games also have standardized rating systems that have been studied (Dogruel & Joeckel, 2013). If a social media platform were to develop and institutionalize a standardized symbology for content warnings, effects stemming from only using a text-based content label may be mitigated.

Even if the symbology for truth scales is absorbed by an online community, truth scales have been criticized for having opaque methodologies or being subjective. Uscinski and Butler (2013) argue that truth scales do not have clearly explained thresholds between their

categories. For example, on PolitiFact, it is unclear what the difference is between a “mostly false” statement and a “pants-on-fire” statement, or in the *Washington Post* fact-checking system, it is unclear what the difference is between three and four Pinocchios.³ In short, the truth scale ratings imply a level of scientific rigor that may not be present in the actual fact-checking system.

4 | THE UNINTENDED EFFECTS OF LABELING

There are additionally several unintended psychological phenomena that should be considered in the context of labeling, namely the illusory truth effect, the continued influence effect, the backfire effect, and the implied truth effect. The illusory truth effect is when the repetition of information strengthens the belief of that information. The continued influence effect is when misinformation continues to influence decision making after a correction has been issued, and the backfire effect theoretically occurs when correcting someone only further increases their belief in the misinformation. Finally, the implied truth effect is when a subset of misinformation is labeled as false, people may perceive unlabeled misinformation as vetted and true. All of these unintended effects that we discuss more in-depth in this section highlight the fact that labeling any piece of information is not straightforward and the practice should be carefully considered.

4.1 | Illusory truth

On balance, the available research literature suggests that content moderators should be careful about directly repeating the misinformation in the absence of a correction or a “false tag” that clearly informs the reader that the information is inaccurate. This is because repeated statements (even those that are false) are more likely to be perceived as being true than newly presented statements (even those that are true). This phenomenon is known as the “illusory truth effect” (Begg et al., 1992; for a review see Brashier & Marsh, 2020). A seminal study was conducted by Hasher et al. (1977), and the phenomenon has been shown to be robust. Furthermore, prior knowledge does not protect against the illusory truth effect. For example, even if people know that Helsinki is the capital of Finland, they will be more likely to endorse that Oslo is the capital if they are repeatedly exposed to the falsehood that “Oslo is the capital of Finland” (Fazio et al., 2015). In other words, the illusory truth can influence judgments, even when people should be able to

draw upon stored knowledge. However, plausibility does serve as a boundary condition for illusory truth where statements with extremely low believability, such as “the Earth is a perfect square,” did not increase perceptions of accuracy when repeated (Pennycook et al., 2018).

Although the exact mechanisms of the illusory truth effect are under debate, it is commonly thought to be due to processing fluency (for a review see Unkelbach et al., 2019). Statements that are repeated are processed more fluently, and more likely to be rated as true. Processing fluency has also been said to lead to rhyming statements more likely to be rated as accurate than non-rhyming statements (McGlone & Tofighbakhsh, 2000), and statements spoken by a native speaker are more likely to be rated as true than those from a second language speaker (Lev-Ari & Keysar, 2010). Well intentioned efforts to correct misinformation may reinforce false beliefs if they repeat the initial piece of misinformation without a sufficiently salient corrective element. The prevalence of the illusory truth effect suggests that the correction needs to be saliently paired with the inaccurate statement.

4.2 | Continued influence

The continued influence effect is where, despite corrections, original misperceptions still influence memory and decision-making processes. The seminal research on the effect is an experiment where participants have presented a story about a warehouse fire (Johnson & Seifert, 1994). The story initially attributes the fire's cause to volatile chemicals and gas cylinders stored in the warehouse. This is later corrected: participants are told that there were never any chemicals or cylinders. While participants often accept the correction, the initial misinformation still influences participants' reasoning. For instance, if they are asked “why there was so much smoke” participants at times report “gas cylinders” as the cause, even when they explicitly acknowledge that there were no gas cylinders in the warehouse. Although the continued influence effect can never be eliminated completely (at the population level), there are several ways it can be attenuated. For example, giving a warning that explains the continued influence may weaken the effect (Johnson & Seifert, 1999). However, even if participants are directly warned about the continued influence effect, the phenomenon is rarely eliminated (Ecker et al., 2010). [Correction added on April 04, 2022, after first online publication: Repeated sentences were removed from this paragraph.]

Relevant to content labeling, the continued influence effect is significantly mitigated when an alternative explanation is given instead of just retracting the

misinformation (Lewandowsky et al., 2020). For example, instead of stating that “the warehouse fire was not caused by gas cylinders,” one could state that “the warehouse fire was not caused by gas cylinders, it was arson.” This allows for the “switching out” of the misinformation for the new factual alternative, plugging the gap in a person's mental model of the situation (Ecker et al., 2010). Future research should investigate how algorithms can provide factual alternatives for specific pieces of misinformation online.

4.3 | Backfire

In general, people's false beliefs can be corrected, and misperceptions can be changed. While some of the early psychological and political science research indicated that a correction could “backfire,” causing that individual to increase belief in the misinformation (Nyhan & Reifler, 2010), the current scientific consensus is that this rarely, if ever, occurs (Lewandowsky et al., 2020). The original Nyhan and Reifler (2010) study suggesting that worldview backfire effects occur, used the case study of weapons of mass destruction (WMDs) in Iraq. The authors found that conservative participants *increased* their belief that WMDs were found in Iraq, when presented with corrective evidence. However, subsequent research has found the backfire effect difficult to replicate (Amazeen et al., 2018; Ecker et al., 2020; Pennycook et al., 2020; Weeks & Garrett, 2014; Wood & Porter, 2019). Failures to replicate the backfire effect indicate that the effect is not a robust empirical phenomenon. While it could be that it is extremely item-, situation-, or individual-specific, it might also reflect that the phenomenon does not exist at all (see Swire-Thompson et al., 2020 for a review).

Some of the difficulties in interpreting findings from the backfire effect literature are due to methodological concerns. For example, few papers report the reliability of their measures. In a meta-analysis, Swire-Thompson et al. (2020) found that 81% of backfire effects were elicited when experiments used single-item measures (i.e., a single question to measure an attitude or belief), which are notoriously unreliable. A subsequent study has shown that backfire effects are indeed strongly associated with reliability (Swire-Thompson et al., 2022). Low reliability could either be attributed to measurement error, or that participants do not know where they stand on certain topics. By improving the methodology of the research concerned with corrections, scholars can have more confidence in their findings and better understand how content moderation, including labels, can be used more effectively. The over-generalization of the early literature that reported backfire effects has had the lasting

effect of hampering actual social media platform policies directed at correcting misinformation (Brodwin, 2020; Porter & Wood, 2020a, 2020b).

4.4 | Implied truth

Although fact-check labels can successfully correct misinformation, corrections rarely appear in isolation in the context of a fast-moving, online information exchange environment (e.g., a Twitter feed). Indeed, users will tend to encounter labeled content among many other posts, photos, videos, advertisements, and various other pieces of information, which can give rise to various unintended interaction effects. One such interaction effect could occur when labeled content appears next to other content without labels. A potential unintended consequence of the typical mixed social media presentation format is the “implied truth” effect, where content *without* a label is more likely to be considered true when presented among content with a “false” or “disputed” label (Pennycook et al., 2020).

Pennycook and colleagues show that when headlines with fact checks and unlabeled headlines are presented together, as they would be in a social media feed, the fact-checks of false headlines do reduce belief in misperceptions. Unfortunately, any unlabeled false headlines are perceived to be more accurate than they would be if encountered in a feed with no labels at all. This finding raises the specter of possibly increasing a user's confidence in misinformation if some, but not all, misinformation is labeled as false or disputed. However, the authors also found that attaching veracity labels to some true headlines “slightly reverses” (Pennycook et al., 2020, p. 4944) the implied truth effect. The implied truth effect, in any case, demands more research to further determine the contours and characteristics of how implied truth works and its effects on moderation policies. This is one place where research on information labeling in other, unrelated areas, may be worth consulting, such as the debates over labeling of genetically modified foods (Messer et al., 2017; Yeh et al., 2019).

5 | THE FUTURE OF CONTENT LABELING

Technology and social media development move quickly and the field of content moderation moves quickly in response. The future of content labeling may look like a combination of various labeling tools and methods. In this section, we highlight two bodies of literature that can contribute to our understanding of what the future of content labeling may look like crowdsourcing and media

literacy. Additionally, we propose a taxonomy of content moderation labeling styles and characteristics to assist in framing future discussions of labeling practice. Crowdsourcing provides social media platforms with a method that may be both democratic and effective at scale. Media literacy speaks to the potential of social media platforms for teaching their users to consume information more critically. Crowdsourcing, combined with media literacy measures and the various labeling considerations highlighted in the taxonomy, may provide a path forward for moderation practitioners.

5.1 | Crowdsourcing

An alternative to both AI-based and claim-specific “artisanal” moderation methods (Caplan, 2018) is the possibility of crowdsourcing moderation, thus drawing upon a large number of users on social media platforms. Pennycook and Rand (2019a) and Allen et al. (2021) found that crowdsourced news source credibility ratings correlate highly with professional fact-checker ratings, suggesting that a crowdsourced approach to labeling may be effective. Pennycook and Rand tested the “wisdom of the crowds” by asking participants to rate their familiarity and trust of 60 news outlets, ranging from mainstream media outlets, to hyper-partisan sources, to outlets posting blatant lies, finding that familiarity plays an important role in differentiating between trustworthy and untrustworthy sources. In Pennycook and Rand (2019a) most participants were unfamiliar with most hyper-partisan and “fake news” sources, and when a participant was unfamiliar with a source, they “overwhelmingly distrusted” (p. 2524) the source. Participants across the political spectrum consistently rated the mainstream sources as more trustworthy than the other two types of news outlets. Using crowdsourcing to up- or down-rank sources based on credibility may be a future possibility for content moderation. For example, since January 2021, Twitter has been experimenting with what they are calling “Birdwatch” (Coleman, 2021), a crowdsourced credibility system, but data on the effectiveness of a system like this is not yet publicly available.

One concern with using crowdsourced ratings is the potential for coordinated behavior by ill-intentioned actors to distort the actual crowdsourced ratings using bots or troll groups (Howard, 2020). Allen et al. (2021) argue that this may be mitigated through careful selection of the crowdsourced participants. However, it is clear that platforms using a crowdsourcing strategy will need mechanisms for catching any coordinated behavior and monitoring the crowdsourced ratings for inauthenticity or unreliability. For example, unreliable behavior may

include the use of coordinated up- or down-voting and flagging to silence dissenting or disenfranchised voices (Crawford & Gillespie, 2016; Tripodi, 2017). Alternatively, there is concern that news outlets could “game” the crowdsourcing system, and it is also not clear what criteria would be used to select the members of the “crowd” who would do the ratings or how the crowd members would be recruited (Pennycook & Rand, 2019a). In any case, while it shows tremendous promise, the research into crowdsourcing moderation is early and Twitter’s experimentation with their Birdwatch pilot program may reveal the method’s efficacy before long.

5.2 | Media literacy

Media literacy is the ability to think critically and personally reflect on news media and one’s news consumption habits (Hobbs, 2010). A central topic in media literacy research is whether the public can distinguish news articles from op-eds, or statements of facts from mere opinion. According to Mitchell et al. (2018), many Americans struggle to make these kinds of distinctions. For example, when quizzed on identifying five factual and five opinion news statements, only 26% of adults were able to correctly identify all five factual statements as fact and only 35% of adults were able to correctly identify all five opinion statements as opinion. One solution would be to clearly label news stories from opinion stories, though this may require changes in the design of how social media presents content. That said, labeling on social media can only go so far when distinguishing news from op-eds because of platform non-interoperability. Opinions and news labels from a news organization may not be evident when they are carried over to social media and shared. Furthermore, labeling is contentious within news organizations themselves, as illustrated by internal arguments in 2020 at the *Wall Street Journal* where the newsroom wanted to better distinguish between news stories and op-eds, but the editorial board pushed back, calling the attempts to better distinguish the two types of stories “cancel culture.”⁴

Another media literacy solution is to label with the intent of teaching “lateral reading” (Wineburg & McGrew, 2017), or look to other sources for more information about the credibility of a source. For example, if someone finds themselves viewing a piece of content from an unfamiliar source, a lateral reading strategy would be to look outside of the source’s website itself to other sources. One might search for the source in a search engine to find information about the source itself, including articles that may discuss reporting practices,

biases, or agendas. Encouraging lateral reading is an educational tool where labels might inform you how you came about your information and why it is being presented to you. An obstacle to lateral reading is search engine optimization (Tripodi, 2017, 2018) where creative domain names or metadata tags (Acker, 2018; Acker & Donovan, 2019) are used to better place a URL in a Google search. Additionally, “data voids,” or where “search terms for which the available data is limited, non-existent, or deeply problematic” (Golebiewski and Boyd, 2018, p. 1) can be taken advantage of by misinformation peddlers (Donovan and Boyd, 2019). In other words, if someone attempts to perform “lateral reading” on an emerging topic or a new source, there may not be enough information available elsewhere for that person to make an informed judgment on credibility.

Early studies suggest that interventions such as labeling may assist in educating people about lateral reading or reminding people to engage in the practice, but it is unclear what exact form the interventions should take (Breakstone et al., 2021; Kohnen et al., 2020). Given that most people do not understand the inner workings of the technology they use, be it TikTok, Twitter, or Google search (Tripodi, 2018), lateral reading labels could facilitate social media learning by teaching people why they are seeing pieces of content. In other words, efforts to teach people social media literacy via labeling could also teach about technology.

Social media labels may even help teach individuals about how to evaluate information online. It has long been proposed that the deliberate use of misinformation could be used as an educational tool (Bedford, 2010). More recently, there have been numerous developments with the practice of inoculation, where individuals are exposed to weakened doses of the misinformation and taught about techniques used by those peddling misinformation (Van der Linden et al., 2017). In other words, instead of shielding people from every false or misleading story, people can be taught the flawed argumentative techniques and persuasive rhetoric used by misleading content such as propaganda, conspiracy theories, or disinformation (Hobbs & McGee, 2014). So-called “inoculation” has been found to reduce the continued influence effect (Cook et al., 2017; Guess et al., 2020) and to boost cognitive resistance to conspiratorial thinking (Banas & Miller, 2013; Lewandowsky & Cook, 2020), including anti-vaccination theories (Jolley & Douglas, 2017). Roozenbeek and van der Linden (2019) show that inoculation need not be taught through traditional pedagogical methods. They found that a game about misinformation was able to surface and enhance misinformation strategies and confer what the researchers call “cognitive immunity” to real misinformation stories. Wikipedia may also serve as a

model for educating users about evidence deficits or a lack of credible sources (Jennings, 2008). For example, providing “citation needed” labels or labels from academic sources signal qualities of the credibility of Wikipedia articles (Rowley and Johnson, 2013). It is therefore conceivable that “educational” labels could effectively instruct users about misinformation strategies disreputable sources tend to implement. This could potentially help individuals not only to recognize that the current post is inaccurate but also to help them sort facts from fiction in the future.

5.3 | Taxonomy for the future

The future of online content labeling can be better informed by clear definitions and categorizations of the different label forms and characteristics. To address the inconsistency of content moderation and to better inform labeling practice, we propose the future development of a detailed labeling taxonomy. A useful labeling taxonomy will have at least two umbrella categories: label type or output, and label factor or input variable. Label type refers to the form of the label, while label factors refer to methodological qualities of the label that the platform applying the label can consider.

Content label types exist on a spectrum of directness, from very direct to very broad. Label types include claim-specific (the most direct), topic-specific, source-specific, sensitivity, manipulated media, and path or virality (the most broad). We detail here preliminary definitions:

- The most direct labels are *claim-specific labels* and are attached to a particular post from a specific account; the label seeks to address the precise statements and information in the message in question.
- *Topic-specific labels* do not have the direct connection between the original claim and counterclaim, but rather may serve as a more broad “get-the-facts” effort to spark awareness of possibly misleading claims and to serve up more contextual information.
- *Source-specific labels* help inform social media users about where the piece of content came from, such as publication source.
- *Sensitivity labels* warn social media users about possibly graphic, sexual, or otherwise upsetting content.
- *Manipulated media labels* warn users that the photo, video, or audio they are viewing or listening to has been modified via, for example, Photoshop (or similar utilities), audio-video editing software, or deep-fake technologies.
- *Path or virality labels* are perhaps the broadest form of intervention and are labels informing users about the

virality of content and why they are seeing the content in their feed.

This list is meant to be only partial, but it seeks to encompass most of the practices applied currently in use by major social media platforms. As the online world evolves to include more forms of multimedia and new media forms such as augmented or virtual reality worlds, more forms of labels will certainly be required.

Label methodologies cover a wide variety of variable factors that can apply to a label that may inform the label type applied and play the role of an input consideration for social media platforms. Characteristics include determination process, tightness of fit, level of intervention needed to apply the label, certainty and timing, internal and cross-platform consistency, repeat offenses, problematic individuals or groups, mislabeling costs, and badge-of-honor payoffs. We propose definitions as follows:

- *Determination process* is the internal (or external in the case of the International Fact Check Network) procedure for determining what kind of label is appropriate for the content.
- *Intervention level* is the degree of severity required from a social media platform in labeling content. For example, content posted during a public health crisis, a civil conflict, or a contested election situation may require orders of magnitude more attention to labeling strategy and more friction-based treatments than more ordinary situations.
- *Certainty and timing* refer to the weighing of early intervention with possible censorship versus late intervention with possible virality.
- *Tightness of fit* is the specificity of any counterclaims or contextual information to the post in question and the level of detail provided to users about the determination process above.
- *Internal consistency* relates to the pattern, wording, and design of prior labels that a given platform has used, while cross-platform consistency is the variation of labeling pattern, wording, and design across social media networks.
- *Repeat offenses* refer to the consideration of changing labeling and moderation based on past conduct by a specific individual or group.
- *Problematic individuals or groups* is the consideration of purposeful targeting of content by particular users, what is sometimes called coordinated inauthentic behavior, and may involve the use of bots or automation.
- *Mislabeling costs* are the potential disadvantages (and externalities) of labeling a piece of content, which can be a problem because algorithms have a reasonable rate of false positives.

- A final characteristic is *badge-of-honor payoffs*, which is the potential for a label to be used for external uses (i.e., “Silicon Valley is censoring me. Donatell!”).

Again, this preliminary taxonomy is meant to be suggestive and not comprehensive. All of these forms of labeling types and factors (inputs and outputs), in addition to others we might have missed, can be further developed into a coherent and exhaustive taxonomy, so that researchers can more fully categorize and understand the wide range of possible uses of labels on social media platforms.

6 | CONCLUSION

Recent research into veracity and context labeling on social media platforms adds to the larger body of labeling literature that covers topics such as media labels, nutrition labels, and hazard warning labels. However, our knowledge of the effects of labels in the context of online misinformation is still largely in a nascent state, and we should expect advancement to be difficult for a variety of reasons. Social media researchers somehow must cope with systemic drift (i.e., the way people use platforms shifts over time), meaning that researchers are left to study past behavior as people and platforms adapt to new events and trends. For example, after the 2016 U.S. presidential election, Facebook made the decision to make its algorithm more focused on content from family and friends rather than news content, so conclusions drawn from research into the effects of Facebook’s algorithms before those changes took place may no longer apply. Scholars are therefore concerned about problems of temporal validity, or the decay of knowledge given the rapid pace of technological change (Munger, 2019). Misinformation research will need to be able to use up-to-date data (ideally provided by platforms through new institutions such as Social Science One, see King & Persily, 2020) to view the online information environment in real-time (HKS Misinformation Review, 2020). However, it is worth noting that efforts so far in this direction have continued to encounter major hurdles, as issues of privacy and anonymization complicate data sharing with independent scholars, and companies such as Facebook have made major errors (Alba, 2021; Levy, 2021; Persily, 2021).

There is also the problem of the “arms race” between content moderators and ill-intentioned actors that compete and adapt to new tactics. For misinformation researchers, current studies are looking at previous misinformation practices, while those who seek to spread misperceptions have often moved on to new tactics,

formats, and methods. Further complicating the moderation tools used by platforms in the arms race is the widespread operation of automated bots on these platforms and the difficulty of separating apart a user’s engagement with misinformation and their forming a belief that misinformation is true. People may be sharing misinformation as a warning, or for socialization, or because they find a story humorous or satirical. In other words, it is difficult to create consistent moderation policies in environments where one person shares a recognized piece of misinformation simply because they think it is funny, while another person shares the very same piece of misinformation with the intent to inform others. In the arms race, these factors give the advantage to those seeking to spread misinformation.

For future research, there are several areas of the content labeling literature where there is a clear lack of consensus: the importance of the source (Berinsky, 2017; Dias et al., 2020; Kim et al., 2019), the effects of graphical elements (Amazeen et al., 2018), and why belief change is not sustained over time (Ecker et al., 2020; Swire, Ecker, & Lewandowsky, 2017), given that many labeling studies measure belief directly after a correction has been presented. In addition to those considerations, while the so-called “backfire effect” (Nyhan & Reifler, 2010) has been found not to be a robust empirical phenomenon (Swire-Thompson et al., 2020; Wood & Porter, 2019), the “implied truth effect” (Pennycook et al., 2020) may be detrimental to the long-term application and success of labeling for fact-check purposes. If fact-checking social media content increases the belief of other pieces of content (including potential misinformation), the platforms need to apply fact-checks consistently. More research may find that implied truth is a major obstacle to labeling for content and reinforce the need for media literacy labeling (Guess et al., 2020) and contextual information labeling (Lorenz-Spreen et al., 2020). [Correction added on April 04, 2022, after first online publication: Reference citation for Swire et al. 2017 was removed from this paragraph.]

The advantage of content labels is that they can move beyond the typical binary of possible moderation methods—removal or downranking. Although we have not taken a stand on the normative issues here, labels may also carry a choice-preserving dimension that better respects the autonomy of users. The literature on such normative and consumer/user welfare-related aspects of labels continues to evolve, and Sunstein (2020), for example, argues that policymakers should focus on a general “human well-being” of society when considering labels. Using a more nimble and nuanced approach to content moderation may facilitate platform governance that is both more ethical and

equitable, allowing for varying approaches to the spread of problematic content and its potential harms. A labeling approach, while no panacea, may aid the health of the information and media space, helping to orient users and citizens while retaining freedoms of expression that are necessary for self-determination and core to robust deliberation in democratic societies.

ORCID

Garrett Morrow  <https://orcid.org/0000-0002-3078-7137>

Briony Swire-Thompson  <https://orcid.org/0000-0001-7464-8940>

Matthew Kopec  <https://orcid.org/0000-0002-6475-4368>

ENDNOTES

- ¹ While this was not the case when corrections were compared with controls (who never saw the misinformation nor correction), the authors suggested that studies with detailed debunking might have had stronger misinformation persistence. Given that studies included in the meta-analysis did not themselves manipulate detail, but were coded as being either having detail or no detail by Chan et al., effect sizes between studies could have differed substantially.
- ² For PolitiFact's methodology, see: <https://www.politifact.com/article/2018/feb/12/principles-truth-o-meter-politifacts-methodology-i/>.
- ³ For *Washington Post's* methodology, see: <https://www.washingtonpost.com/politics/2019/01/07/about-fact-checker/>.
- ⁴ See Trachtenburg (2020) where the newsroom of the *Wall Street Journal* requested clearer distinctions between them and the editorial board, which pushed back (see WSJ Editorial Board, 2020).

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